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Efficient Multi-Agent Optimization of Optical Power in S+C+L-Band Systems

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Abstract *We propose an AI Agent tailored for link power management in multi-band systems. In S+C+L band span-level study, the agent efficiently solves various optimization objectives. In network-wide evaluation, it delivers 689.0 Tbps gain in total allocated traffic with merely 303 average interactions per power profile.©2026 The Author(s)*

Introduction

Multi-band transmission (MBT) offers a potential solution to expand the capacity of optical fiber networks [1]. In recent years, the S-band emerged as a viable option to further expand system bandwidth [2]. In S+C+L-band systems, signal power evolution is significantly impaired by amplifier gain ripples, noise figure (NF) variations, and inter-channel stimulated Raman scattering (ISRS). As a result, both the quality of transmission (QoT) uniformity and the overall system throughput are severely compromised. Previous studies have investigated power optimization methods to enhance the performance of multi-band transmission systems [3–8].

Enhancing the capacity of multi-band transmission system can be modeled as a power optimization problem [9], and power optimization is executed on the model-based or model-free way in previous work. Model-based methods [3, 5, 6] optimize the power depending on the models for QoT estimation. However, these methods are often limited in achieving optimal power configuration owing to parameter uncertainties and imperfect modeling [10]. In contrast, model-free methods [7, 8, 11] optimize link power via direct interactions with the environment, regarding multi-band transmission systems as black boxes. Nevertheless, such approaches demand a considerable number of iterative interactions with the transmission system to converge [12].

Recently, the rapid advancement of Large Language Models (LLMs) has spurred the emergence of LLM-based AI agents [13]. Furthermore, such agents have demonstrated capabilities in solving optimization problems [14], making them promising for multi-band transmission system power optimization.

In this paper, Power-Optimization (PO) Agent is specifically designed to optimize link power in multi-band transmission systems, targeting diverse optimization objectives: (i) maximizing total capacity [11]; (ii) flattening the received generalized signal-to-noise ratio (GSNR) [5]; (iii) flattening the received optical signal-to-noise ratio (OSNR) [15]; and (iv) equalizing the received power [3]. During the optimization process, the agent dynamically adopts different optimization strategies based on its interactions with the multi-band system. In the S+C+L-band single span transmission, the proposed agent outperforms the Genetic Algorithm (GA) [8] and Particle Swarm Optimization (PSO) [11] across all objectives under limited interaction budgets. Furthermore, progressive traffic analysis on the Italy network topology demonstrates that PO agent supports a significant enhancement

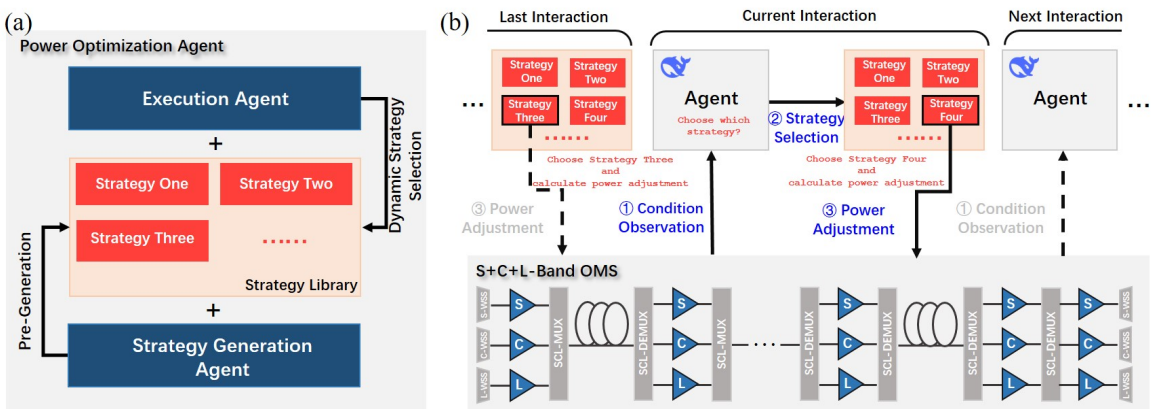

Fig. 1: Structure and workflow of the Power Optimization Agent: (a) agent architecture; (b) operational workflow.

of 689.0 Tbps in total allocated traffic with higher efficiency.

Link power optimization with AI Agent

Fig.1(a) depicts the architecture of the proposed PO Agent for optical power optimization. Built on a multi-agent framework [16], the system consists of two collaborative sub-agents: the execution agent and the strategy generation agent.

The strategy generation Agent is responsible for constructing and maintaining a strategy library. Each entry in this library is a specialized function that outputs link power adjustments, tailored specifically for optical power optimization. By replacing complex numerical calculations with these pre-defined strategies, PO Agent effectively overcomes the inherent limitations of LLMs in precise numerical computation [17].

The execution agent interacts with the multi-band transmission system to adjust link power during the optimization process. First, it observes the current system condition and then selects an appropriate strategy from the pre-defined library based on these observations. Once the strategy calculates the required power adjustment, the system updates the link power accordingly, initiating the next interaction cycle with a new observation.

Fig.1(b) illustrates the detailed operational workflow of the PO Agent. In this framework, the execution agent dynamically retrieves optimal strategies from a library curated by the strategy generation agent. The optimization process continues until the PO Agent terminates autonomously or reaches a pre-set maximum number of interactions.

The objectives of link power optimization are embedded within the PO Agent as system prompts, including maximizing total capacity, flattening the received GSNR and OSNR, and equalizing received power. At the commencement of the optimization, the PO Agent parses the user's requirements and initiates the optimization process. Concurrently, workflow prompts and action rules are embedded within the agent to enhance the reliability of the optimization.

Built upon the LangChain framework [18], the PO Agent employs the ReAct paradigm [19] and leverages DeepSeek-V3.2 [20] as its underlying language model.

Span-Level Optimization

In the span-level study, the launch power of each channel is optimized over an 80 km span of ITU-T G.652D standard single-mode fiber (SSMF) transmission, with the aim of achieving four objectives: maximizing total capacity, flattening the received GSNR and OSNR, and equalizing received power. According to the ITU-T G.694.1 standard [21], the frequency ranges for these bands are defined as 185.975 THz ~190.775 THz, 191.275 THz ~196.075 THz, and 196.575 THz ~206.175 THz, respectively. Each band operates on the ITU-T 50 GHz WDM grid, with transceivers configured for a symbol rate of 32 GBaud. Consequently, 384 channels (96 channels in L-band, 96 channels in C-band, and 192 channels in S-band) are employed and optimized in total for S+C+L-band transmission in this study. The muti-band transmission is simulated with GNPy [22], which considers the amplifier gain ripples, NF variations, Kerr nonlinearity and ISRS.

Meanwhile, GA [8] and PSO [11] are employed as baseline algorithms for performance comparison. For both methods, the maximum interaction budget is set to 250 and the optimization process is repeated for 5 times. Additionally, ideal flexible transceivers are assumed, enabling continuous bit-rate adaptation based on the available GSNR [4]. To enhance convergence speed of GA and PSO, the adjacent channel power approximation [11] is incorporated into the optimization process, with the number of adjacent channels set to 40. On the other hand, PO Agent directs observe and

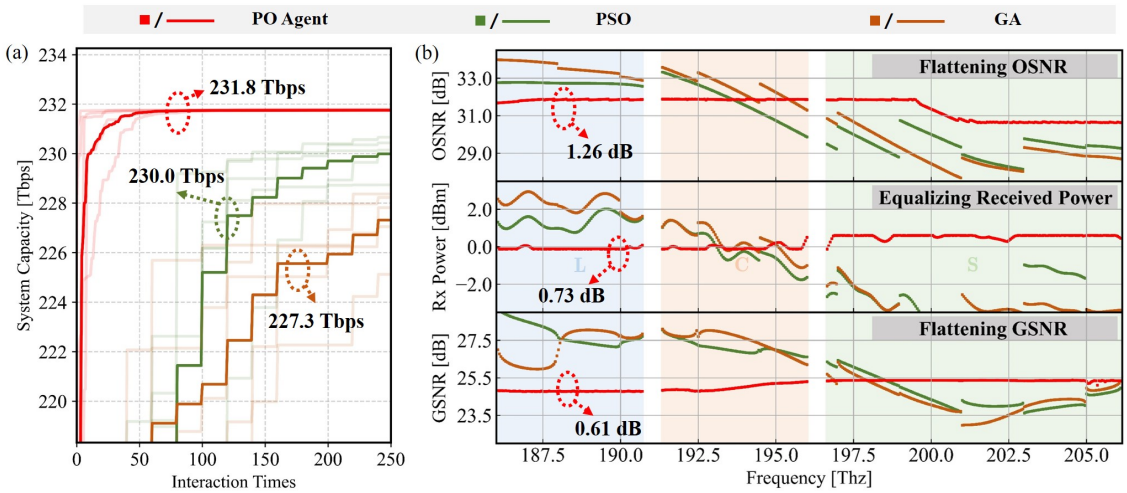

Fig. 2: Span-level optimization results: (a) Total capacity as a function of interaction time; (b) Spectral profiles of the received OSNR (top), power (middle), and GSNR (bottom) after optimization.

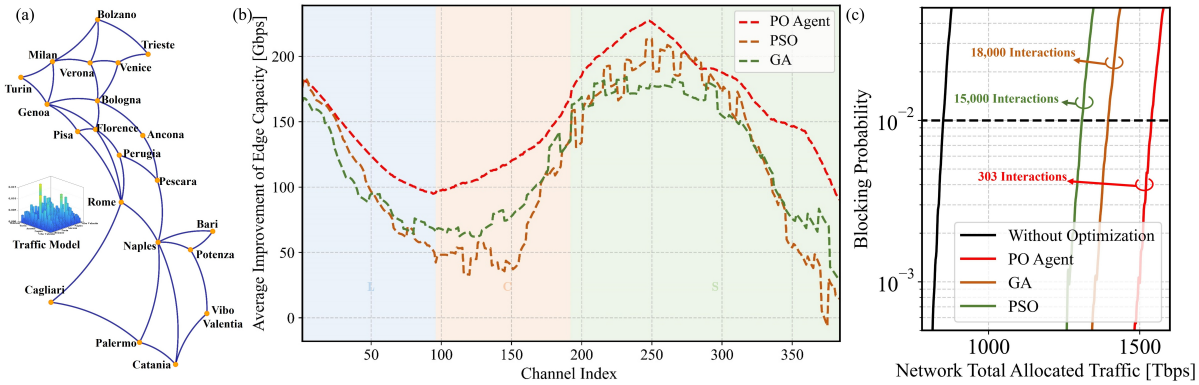

Fig. 3: Network-wide scenario and optimization results: (a) Topology of the Italy network alongside the joint probability density function (PDF) of the population-based traffic model; (b) Average improvement in edge capacity across different algorithms; and (c) Total allocated traffic versus blocking probability.

adjusts the launch power of all channels.

Fig. 2(a) illustrates the total capacity of the 80-km S+C+L-band transmission system as a function of interaction time. As the results show, the total capacity increases rapidly within the first 50 interactions for the PO Agent, whereas PSO and GA converge at a slower rate. Benefiting from this rapid convergence, the PO Agent achieves an average maximum capacity of 231.8 Tbps after 250 interactions, compared to 230.0 Tbps and 227.3 Tbps for PSO and GA, respectively. Fig. 2(b) presents the average optimized results for the remaining targets. After optimization by the PO Agent, the average ripples in received OSNR, power, and GSNR are 1.26 dB, 0.73 dB, and 0.61 dB, respectively. In contrast, the ripples obtained with the other optimization methods all exceed 4 dB.

Network-Wide Optimization

We also compare the PO Agent with PSO and GA over the Italy network [23]. As shown in Fig. 3(a), the Italy network has 21 nodes and 35 links, with an average nodal degree of 3.3, average distance between nodes of 239 km, and maximum link length of 617 km. We approximate the real topology with SSMF of 80 km per span, and suppose that the wavelength-selective switch (WSS) for all band is set after every 3 span transmission for power adjustment. Consistent with the span-level study, network transmission utilizes 384 channels across the S, C, and L bands.

Network power optimization employs a span-by-span strategy utilizing the Local Optimization Global Optimization (LOGO) algorithm [4]. Additionally, the launch power and attenuation are optimized sequentially [5] to shape the power profile (PP). To evaluate performance under various power optimization methods, we adopt the Statistical Network Assessment Process (SNAP) [24], driven by a population-based non-uniform traffic model. The joint probability density function (JPDF) of this non-uniform model, which determines the frequency of requests between each node pair, is derived from the population statistics [3]. We set the Monte Carlo iteration count N_{mc} to 5000 to ensure stable metric convergence. Routing follows a k-shortest path algorithm ($k = 8$), with First-Fit (FF) wavelength assignment (WA) used during progressive traffic analysis to capture both dynamic and static performance metrics.

For network power optimization, we impose no constraints on the maximum number of interactions for the PO Agent; instead, it determines when to terminate the current optimization process autonomously. In contrast, for PSO and GA, the maximum number of interactions is set to 15,000 and 18,000, and the number of adjacent channels is configured to 10 to enhance optimization performance.

Fig. 3(b) illustrates the average improvement in edge capacity across all channels for every network link. The results demonstrate that the PO Agent achieves the highest average improvement for nearly all channels compared to baseline methods. Fig. 3(c) depicts the relationship between the total allocated traffic enhancement and the blocking probability (BP) threshold, set at 0.01. Near this threshold, the PO Agent delivers the most significant improvement, achieving a total throughput of 689.0 Tbps. In comparison, PSO and GA achieve improvements of 458.1 Tbps and 546.0 Tbps, respectively. Additionally, the PO Agent demonstrates superior efficiency, requiring an average of only 303 interactions per PP to converge.

Conclusion

In this paper, the PO Agent is specifically designed to optimize link power in multi-band transmission systems, significantly enhancing optimization efficiency. In a span-level study across S+C+L bands, the PO Agent outperforms both PSO and GA across various optimization objectives. Furthermore, in a network-wide evaluation using the real-world Italy topology, the PO Agent achieves a significant enhancement of 689.0 Tbps in total allocated traffic, requiring an average of only 303 interactions per PP.

Acknowledgements

One extra page is allowed so that acknowledgements and references can be given in full length.

References

[1] A. Ferrari et al., "Assessment on the achievable throughput of multi-band itu-t g.652.d fiber transmission systems", *Journal of Lightwave Technology*, vol. 38, no. 16, pp. 4279–4291, 2020. DOI: 10.1109/JLT.2020.2989620.

[2] S. Escobar-Landero, A. Lorences-Riesgo, X. Zhao, Y. Frignac, and G. Charlet, "S+c+l high-capacity transmission systems: Challenges and opportunities", *Journal of Lightwave Technology*, vol. 42, no. 12, pp. 4260–4270, 2024. DOI: 10.1109/JLT.2024.3401540.

[3] F. Arpanaei et al., "Synergizing hyper-accelerated power optimization and wavelength-dependent qot-aware cross-layer design in next-generation multi-band eons", *IEEE Journal on Selected Areas in Communications*, vol. 43, no. 5, pp. 1840–1855, 2025. DOI: 10.1109/JSAC.2025.3543528.

[4] B. Correia et al., "Power control strategies and network performance assessment for c+l+s multiband optical transport", *Journal of Optical Communications and Networking*, vol. 13, no. 7, pp. 147–157, 2021. DOI: 10.1364/JOCN.419293.

[5] Y. Zhang et al., "Optical power control for gsnr optimization based on c+l-band digital twin systems", *Journal of Lightwave Technology*, vol. 42, no. 1, pp. 95–105, 2024. DOI: 10.1109/JLT.2023.3303783.

[6] X. Pang et al., "Digital twin-assisted optical power allocation for flexible and customizable snr optimization", in *2022 Optical Fiber Communications Conference and Exhibition (OFC)*, 2022, pp. 1–3.

[7] M. Ran, M. Gong, Q. Yin, M. Luo, T. Huang, and X. Li, "Optimization strategy for power control in c+l+s band transmission using particle swarm optimization", pp. 319–323, 2025. DOI: 10.1109/WOCC63563.2025.11082171.

[8] N. A. Shevchenko, S. Nallaperuma, and S. J. Savory, "Ultra-wideband information throughput attained via launch power allocation", in *2021 International Conference on Optical Network Design and Modeling (ONDM)*, 2021, pp. 1–3. DOI: 10.23919/ONDM51796.2021.9492399.

[9] I. Roberts, J. M. Kahn, and D. Boertjes, "Convex channel power optimization in nonlinear wdm systems using gaussian noise model", *Journal of Lightwave Technology*, vol. 34, no. 13, pp. 3212–3222, 2016. DOI: 10.1109/JLT.2016.2569073.

[10] Q. Zhuge et al., "Building a digital twin for intelligent optical networks [invited tutorial]", *Journal of Optical Communications and Networking*, vol. 15, no. 8, pp. C242–C262, 2023. DOI: 10.1364/JOCN.483600.

[11] D. Semrau, E. Sillekens, P. Bayvel, and R. I. Killey, "Modeling and mitigation of fiber nonlinearity in wideband optical signal transmission [invited]", *Journal of Optical Communications and Networking*, vol. 12, no. 6, pp. C68–C76, 2020. DOI: 10.1364/JOCN.382267.

[12] Y. Song, Q. Fan, C. Lu, D. Wang, and A. P. T. Lau, "Efficient three-step amplifier configuration algorithm for dynamic c+l-band links in presence of stimulated raman scattering", *Journal of Lightwave Technology*, vol. 41, no. 5, pp. 1445–1453, 2023. DOI: 10.1109/JLT.2022.3223919.

[13] Z. Xi et al., *The rise and potential of large language model based agents: A survey*, 2023. arXiv: 2309.07864 [cs.AI]. [Online]. Available: https://arxiv.org/abs/2309.07864.

[14] S. Liu, C. Gao, and Y. Li, "AgentHPO: Large language model agent for hyper-parameter optimization", in *The Second Conference on Parsimony and Learning (Proceedings Track)*, 2025. [Online]. Available: https://openreview.net/forum?id=HU3yfXcoKU.

[15] L. Rapp and M. Eiselt, "Optical amplifiers for multi–band optical transmission systems", *Journal of Lightwave Technology*, vol. 40, no. 6, pp. 1579–1589, 2022. DOI: 10.1109/JLT.2021.3120944.

[16] T. Guo et al., "Large language model based multi-agents: A survey of progress and challenges", in *Proceedings of the Thirty-Third International Joint Conference on Artificial Intelligence, IJCAI-24*, K. Larson, Ed., Survey Track, International Joint Conferences on Artificial Intelligence Organization, Aug. 2024, pp. 8048–8057. DOI: 10.24963/ijcai.2024/890. [Online]. Available: https://doi.org/10.24963/ijcai.2024/890.

[17] H. Li et al., "Exposing numeracy gaps: A benchmark to evaluate fundamental numerical abilities in large language models", in *Findings of the Association for Computational Linguistics: ACL 2025*, W. Che, J. Nabende, E. Shutova, and M. T. Pilehvar, Eds., Vienna, Austria: Association for Computational Linguistics, Jul. 2025, pp. 20 004–20 026, ISBN: 979-8-89176-256-5. DOI: 10.18653/v1/2025.findings-acl.1026. [Online]. Available: https://aclanthology.org/2025.findings-acl.1026/.

[18] O. Topsakal and T. C. Akinci, "Creating large language model applications utilizing langchain: A primer on developing llm apps fast", *International Conference on Applied Engineering and Natural Sciences*, vol. 1, pp. 1050–1056, Jul. 2023. DOI: 10.59287/icaens.1127.

[19] S. Yao et al., *React: Synergizing reasoning and acting in language models*, 2023. arXiv: 2210.03629 [cs.CL]. [Online]. Available: https://arxiv.org/abs/2210.03629.

[20] DeepSeek-AI et al., *Deepseek-v3.2: Pushing the frontier of open large language models*, 2025. arXiv: 2512.02556 [cs.CL]. [Online]. Available: https://arxiv.org/abs/2512.02556.

[21] H. Luo, J. Lu, Z. Huang, C. Yu, and C. Lu, "Optimization strategy of power control for c+l+s band transmission using a simulated annealing algorithm", *Opt. Express*, vol. 30, no. 1, pp. 664–675, Jan. 2022. DOI: 10.1364/OE.439635. [Online]. Available: https://opg.optica.org/oe/abstract.cfm?URI=oe-30-1-664.

[22] A. Ferrari et al., "Gnpy: An open source application for physical layer aware open optical networks", *Journal of Optical Communications and Networking*, vol. 12, no. 6, pp. C31–C40, 2020. DOI: 10.1364/JOCN.382906.

[23] R. Matzner et al., "Topology bench: Systematic graph-based benchmarking for core optical networks", *Journal of Optical Communications and Networking*, vol. 17, no. 1, pp. 7–27, 2025. DOI: 10.1364/JOCN.534477.

[24] V. Curri, M. Cantono, and R. Gaudino, "Elastic all-optical networks: A new paradigm enabled by the physical layer. how to optimize network performances?", *Journal of Lightwave Technology*, vol. 35, no. 6, pp. 1211–1221, 2017. DOI: 10.1109/JLT.2017.2657231.